# FINITENESS OF THE COHOMOLOGY OF TORI AND OF FINITE MULTIPLICATIVE-TYPE GROUP SCHEMES OVER A CURVE

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*Allowed standard prerequisites (editorial).*

- Galois cohomology of a torus split by a finite Galois covering, via the Hochschild-Serre spectral sequence. Used inside the counted proof at lines 578-591. Produces the group G = Gal(V/U) and the exact sequence 0 -> H^1(G, T(V)) -> H^1(U, T) -> H^1(V, T_V).
- Finiteness of the ideal class group and Dirichlet's unit theorem over a global field. Used at lines 582 and 593-595. Makes H^0(V, G) of finite type and H^1(V, G) of finite type (finite when U is not X).
- Artin-Mazur-Milne duality for fppf cohomology. Cited at lines 622-623. Transfers the finiteness of H^1(U, N) to H^2_c(U, N^D).

**Lemma 1.1.** (a) *Let $\mathscr{T}$ be a torus over $U$. Then $H^0(U, \mathscr{T})$ and $H^1(U, \mathscr{T})$ are of finite type. If $U \neq X$, then $H^1(U, \mathscr{T})$ is finite.*

(b) *Let $\mathscr{N}$ be a finite group scheme of multiplicative type over $U$. Then $H^1(U, \mathscr{N})$ and $H^2_c(U, \mathscr{N}^D)$ are finite.*

*Proof.* (a) Let $V$ be an étale and finite connected Galois covering of $U$ such that $\mathscr{T}_V := \mathscr{T} \times_U V$ is isomorphic to $\mathbf{G}_m^r$ for some non negative integer $r$. The group $\mathscr{T}(V) \simeq H^0(V, \mathbf{G}_m)^r$ is of finite type by Dirichlet's theorem on units. Therefore $H^0(U, \mathscr{T}) \subset H^0(V, \mathscr{T}) = \mathscr{T}(V)$ is also of finite type.

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Set $G = \mathrm{Gal}(V/U)$. Then the Hochschild-Serre spectral sequence provides an exact sequence

$$0 \to H^1(G, \mathscr{T}(V)) \to H^1(U, \mathscr{T}) \to H^1(V, \mathscr{T}_V).$$

Since $\mathscr{T}_V \cong \mathbf{G}_m^r$, the group $H^1(V, \mathscr{T}_V) \cong (\mathrm{Pic}\, V)^r$ is of finite type (resp. finite if $U \neq X$) by finiteness of the ideal class group of a global field. As $\mathscr{T}(V)$ is of finite type and $G$ is finite, the group $H^1(G, \mathscr{T}(V))$ is finite by [20, Chap. VIII, Cor. 2]. Thus $H^1(U, \mathscr{T})$ is of finite type (resp. finite if $U \neq X$) as well.

(b) We can assume (by ([16, Lem. III.8.9] and [8, Cor. 4.9]) that $U \neq X$. Since every finitely generated Galois module is a quotient of a torsion-free and finitely generated Galois module, the assumption that $\mathscr{N}$ is of multiplicative type implies (by [22, X, Prop. 1.1]) that there is an exact sequence of $U$-group schemes

$$0 \to \mathscr{N} \to \mathscr{T}_1 \to \mathscr{T}_2 \to 0,$$

where $\mathscr{T}_1$ and $\mathscr{T}_2$ are $U$-tori. Therefore, there is an exact sequence of abelian groups

$$H^0(U, \mathscr{T}_2) \to H^1(U, \mathscr{N}) \to H^1(U, \mathscr{T}_1).$$

By (a), we know that $H^1(U, \mathscr{T}_1)$ is finite. Let $n$ be the order of $\mathscr{N}$; then the map $H^0(U, \mathscr{T}_2) \to H^1(U, \mathscr{N})$ factorizes through a map $H^0(U, \mathscr{T}_2)/n \to H^1(U, \mathscr{N})$. But $H^0(U, \mathscr{T}_2)/n$ is finite because $H^0(U, \mathscr{T}_2)$ is of finite type by (a). Finally $H^1(U, \mathscr{N})$ is finite. The finiteness of $H^2_c(U, N^D)$ follows by Artin-Mazur-Milne duality ([8, Th. 1.1]). □

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